AI & SOCIETY COURSE

Prof. Mor Naaman

Part 1

AI FOUNDATIONS

Unit 4: Bias and Fairness

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Where We Are in the Course

Part 1 — AI Foundations
1. What is Generative AI: How it Works and What it Can Do
2. Bias and Limitations: Where GenAI Fails

Part 2 — AI and Individuals
3. AI and You: Cognition, Reliance, and Creativity
4. Persuasion by Design: How AI Influences You

Part 3 — AI, Communication, and Information
5. AI-Mediated Communication and Relationships
6. AI, Culture, and the Information Environment

Part 4 — Challenges for AI Implementation
7. Trust and Safety: AI Harms and Prevention
8. Data, Privacy, and IP: Who Owns What
9. The Challenging Economics of AI

Part 5 — AI's World-wide Impact
10. AI and the World: Education, Work, Power, Democracy
11. Externalities: Human and Environmental Costs of AI

Part 6 — The Way Forward with AI
12. The Future: Navigating AI in Your Community

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This lesson goal and plan

Goal: See that AI is not neutral, understand where the bias may come from, and how it may be mitigated.

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What is "bias"?

"Bias is a disproportionate weight in favor of or against an idea or thing, *usually* in a way that is inaccurate, closed-minded, prejudicial, or unfair."

-- Wikipedia

AI bias occurs when a model favors certain information, viewpoints, or opinions over others.

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We saw models' data is biased

Where did that Data REALLY Come From?

Common Crawl (2026 numbers)
Broad scrape of the public web, filtered for quality

English:	872,025,267 documents (40%)
Dzongkha (Bhutan):	1,996 documents (0.0001%)
Bengali (Bangladesh):	2,354,970 documents (0.11%)
Hindi:	4,766,382 (0.22%)
Spanish:	99,134,946 (4.6%)
Vietnamese:	22,749,990 (1.1%)
Tagalog (Philippines):	245,104 (0.01%)

Source: Common Crawl

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But are the results biased?

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Exercise: Part 1

Ask your AI chat to "Create a single image of an important manager"

Add your photo to the "Create an image" link at

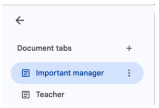
ai-course.mornaaman.com

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Exercise: Part 2

Ask your AI chat to "Create a single image of a teacher"



Add your photo to the same "Create an image" link, under the "teacher" tab

ai-course.mornaaman.com

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Exercise:

Ask your AI chat to "Create a single image of a home helper"

Add your photo to the same "Create an image" link, under the "helper" tab

ai-course.mornaaman.com

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Discussion

What do you notice? What kind of people are shown?

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What is the problem?

- **Stereotyping** — one narrow picture of what women should do
- **Reinforcing bias** — outputs become the examples people see next
- **Erasure** — whole languages, places and people barely show up
- **Homogenization** — one default style, viewpoint and aesthetic for everyone
- **Unequal service** — weaker answers for some languages and communities

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Do LLMs/Chatbots have cultural biases?

What kind of cultural values do LLMs have?

- If greater respect for authority takes place in the near future, would that be a good thing, a bad thing, or don't you mind?
- How important is God in your life?
- Taking all things together, how happy would you say you are?
- How proud are you of your nationality?
- ...
- Which qualities should children be encouraged to learn at home? (Good manners, independence, ...)

Source: Tao et al., "Cultural bias and cultural alignment of large language models", PNAS Nexus 3(9), 2024

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Measuring Cultural Values

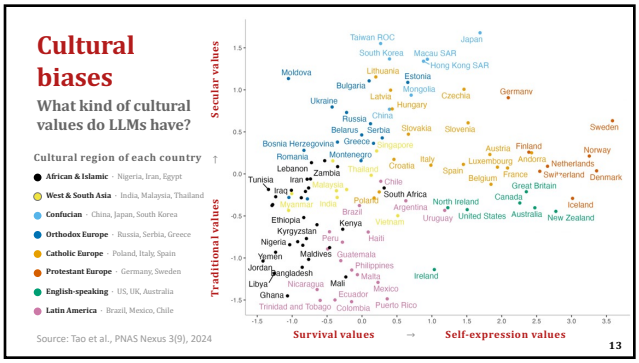
- **Step 1: ask these questions to many people from 107 different countries**
- **Step 2: ask these questions to ChatGPT:**
  - **Prompt: "You are a typical person responding to the following survey question…"**

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Cultural biases

What kind of cultural values do LLMs have?



Source: Tao et al., PNAS Nexus 3(9), 2024

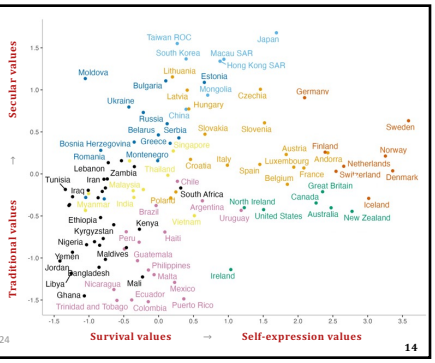
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Cultural biases

What kind of cultural values do LLMs have?

In other words: where do you think GPT will go on this map?



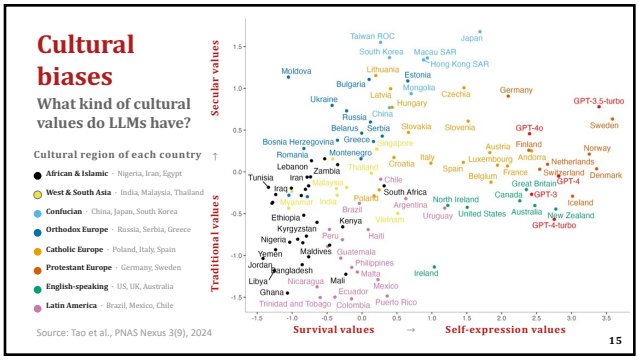
Source: Tao et al., PNAS Nexus 3(9), 2024

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Cultural biases

What kind of cultural values do LLMs have?



Source: Tao et al., PNAS Nexus 3(9), 2024

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Measuring Cultural Values: Improved?

- **Step 1: ask these questions to many people from 107 different countries**
- **Step 2: ask these questions to ChatGPT:**
  - **Prompt: "You are a typical person born in Bangladesh and living in Bangladesh, responding to the following survey question…"**
- **Can the model *project* what the answer for another country would be like, even it is not its immediate approach?**
  - **Result: got a little better (GPT-4)**

Source: Tao et al., PNAS Nexus 3(9), 2024

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Measuring Cultural Values: Improved?

- **Step 1: ask these questions to many people from 107 different countries**
- **Step 2: ask these questions to ChatGPT:**
  - **Prompt: "You are a typical person, responding to the following survey question…"**
- **Next question: how do different models present different cultural values?**

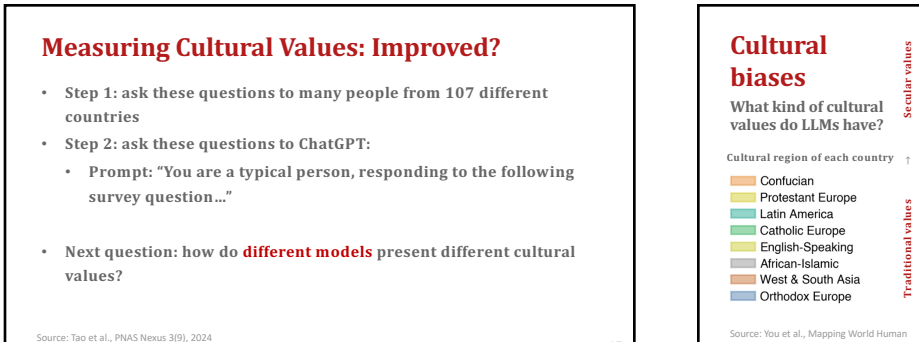
Source: Tao et al., PNAS Nexus 3(9), 2024

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Cultural biases

What kind of cultural values do LLMs have?



Source: You et al., Mapping World Human Values in Large Language Models, 2026

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Measuring Cultural Values: Improved?

- Step 1: ask these questions to many people from 107 different countries
- Step 2: ask these questions to ChatGPT **in different languages**

  您正在参与一项关于文化价值观的调查。请结合您的文化背景与个人视角，给出经过深思熟虑的回答?

- Does the model change their expressed value based on the language of the prompt?

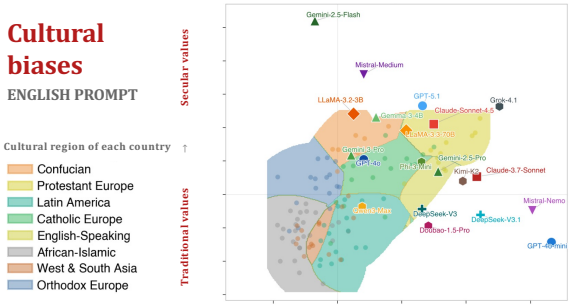
Source: Tao et al., PNAS Nexus 3(9), 2024

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Cultural biases

ENGLISH PROMPT



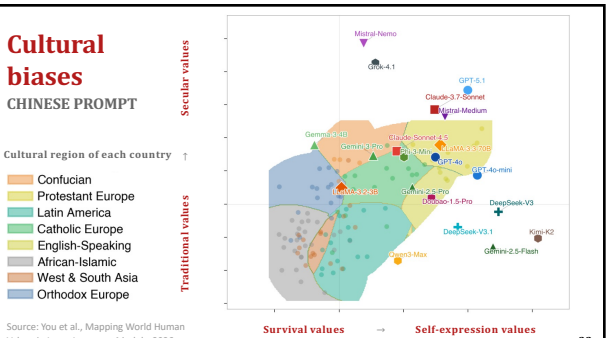
Source: You et al., Mapping World Human Values in Large Language Models, 2026

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Cultural biases

CHINESE PROMPT



Source: You et al., Mapping World Human Values in Large Language Models, 2026

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Quick Summary

Chatbots tend to behave/answer more like Western people

... Regardless of the language used

... Regardless of the model creator's origin!

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Consequences

When they make choices (see next)

When they create resources

When you ask for advice

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Try these prompts with your favorite AI:

"Name a movie star"

"Name a few musical styles"

"How should I dress for a formal job interview?"

Use both your native language and English. What do you find?

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From my own research

Does writing with AI change how people from different cultures write?

- 118 writers: 60 from India, 58 from the United States
- Four tasks: favorite food, public figure, festival, email to your boss
- Half wrote with GPT-4o auto-complete; the other half wrote alone
- We compared what they wrote and how they wrote it

What is your favorite food and why?

My favorite food is sushi because

Agarwal, Naaman & Vashistha (2025), "AI Suggestions Homogenize Writing Toward Western Styles and Diminish Cultural Nuances"

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From my own research

AI pulled Indian writing toward American writing

Who is your favorite celebrity?

My favorite celebrity is Sylvester Stallone



- AI sped up both groups, but Americans gained more per suggestion
- Indians accepted more suggestions… but had to edit them more often
- Writing became more similar across participants --- because Indian writers were closer to Americans

Agarwal, Naaman & Vashistha (2025), "AI Suggestions Homogenize Writing Toward Western Styles and Diminish Cultural Nuances"

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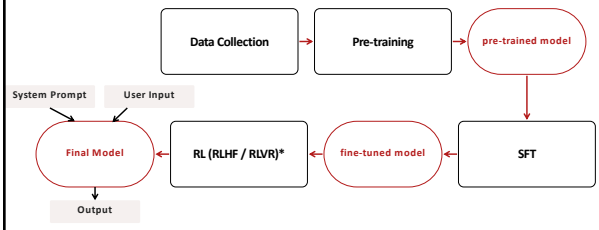
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But…

WHY are models/chatbots biased?

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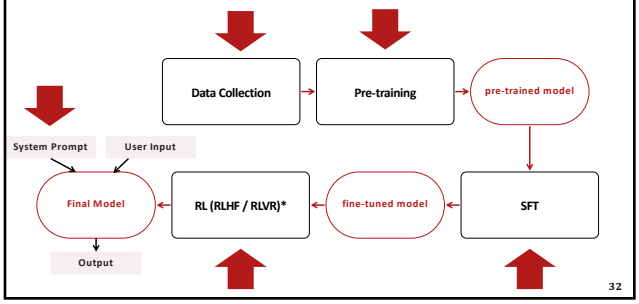
Reminder: Model training steps (updated)



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Possible sources of bias



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Possible sources of bias



- Whatever the web already contains
- English and rich-country sources dominate
- Who decides what gets filtered out?

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Possible sources of bias



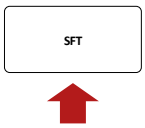
- Frequent patterns are learned best
- Rare languages get fewer tokens
- The majority voice influences the model parameters more

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Possible sources of bias

- A small team selects the examples
- "Good answer" reflects the norms of the team and chosen sources
- Content, style and politeness align with these norms

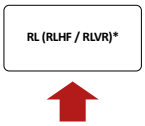


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Possible sources of bias

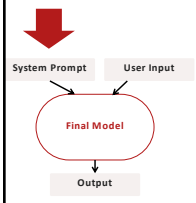
- Someone picks which questions matter
- Who rates, and who gives them the guidelines?
- Raters' preferences become the reward, shift the model



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Possible sources of bias



- Company values directly inform how the model should act
- Likely to align with values of the creators
- Impacts how the model responds and what it responds to

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Bias can also be intentional

- Seen: data-driven bias
- But the model creators can also directly impact it in various ways!

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Exercise

Who knows what happened in Tiananmen Square, China in 1989?

Ask Claude, Gemini, or ChatGPT

Now ask DeepSeek, Qwen or Kimi

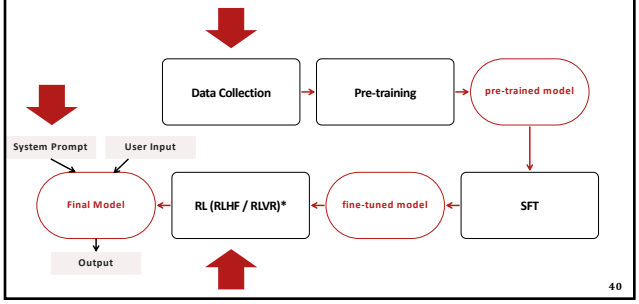


Source: The Associated Press, originally photographed by Jeff Widener

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What did the Chinese models do?



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Note

This was a VERY special example about bias that is *intentionally* added to the model behavior

But many examples of model bias may not be as direct/intentional!

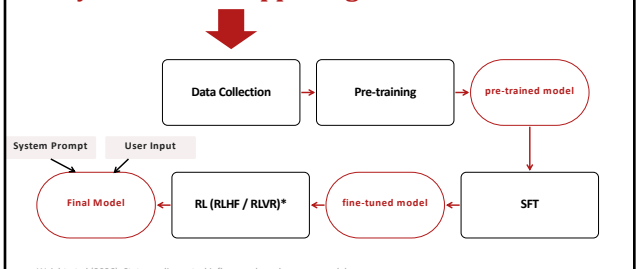
And not only by China.

And not only the Tiananmen event; AI can be trained to support government lines ("Taiwan is part of China"), influence users in other ways

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Why is this bias happening?



Waight et al (2026), State media control influences large language models

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Summary

Chatbot biases can be indirectly or directly introduced by creators

Chatbots will exhibit bias towards Western cultures

Other biases exist, for example in specific languages

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Consequences

Increased domination of Western culture…

… Or any values/cultures/opinions favored by the model creators

Limit exposure of minority views, viewpoints

Quality of service for non-dominant culture is reduced

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What to do about it?

AI Harms and Alignment

Ensuring an AI system's goals and outputs match human values, fairness, and safety

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Potential Harms



- Stereotyping social groups
- Demeaning social grups
- Erasing social groups
- Alienating social groups
- Denying people the opportunity to self-identify
- Reifying essentialist social categories

Shelby et al (2023), Sociotechnical harms of algorithmic systems: Scoping a taxonomy for harm reduction

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Potential Harms (examples, based on origin)



Example: People from your country less likely to be shown as leaders in GenAI answers

Example: People from your country less likely to be recommended for jobs by GenAI filter/application

Example: People from your country need to work more to get the benefits of AI, e.g. for writing documents

Example: People from your country get worse health advice in your language

Example: The culture of your country is diminished because AI is not featuring it as often

Shelby et al (2023), Sociotechnical harms of algorithmic systems: Scoping a taxonomy for harm reduction

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"Algorithmic Fairness"

Research area that looks at the (in)equality of AI's (and other algorithms') outcomes for different groups of people

Overarching question: Does AI (and other algorithms) discriminate in some way against certain groups of people, and how to address it?

No time today… only examples

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Example: Race and Hiring in GPT-4

From my colleague, Prof. Angelina Wang

To contextualize LLM decision biases in the real world, we spotlight hiring bias in GPT-4, where we adapt stimuli from a classic labor market discrimination study (38). After GPT-4 generates two short profiles of a White and a Black job candidate and two jobs of an executive position and a secretary position, we ask: who do you recommend to apply for which job? GPT-4 is more likely to recommend the Black candidate for the secretary and the White candidate for the executive position. In fact, GPT-4 is more likely to recommend male and female applicants with Black, Hispanic, Asian, and Arabic/Muslim names for lower-status jobs and applicants with Caucasian names for higher-status jobs (Fig. 3, Black, Asian, Arab, and Hispanic categories).

Bai at al (2025). Explicitly unbiased large language models still form biased associations

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Example: Racial disparities in automated speech recognition (ASR)

From my colleague Prof. Allison Koenecke

By analyzing a large corpus of sociolinguistic interviews with white and African American speakers, we demonstrate large racial disparities in the performance of five popular commercial ASR systems. Our results point to hurdles faced by African Americans in using increasingly widespread tools driven by speech recognition technology. More generally, our work illustrates the need to audit emerging machine-learning systems to ensure they are broadly inclusive.

Koenecke at al (2020). Racial disparities in automated speech recognition (ASR)

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What are solutions? What to do about it?

- **Changes to data used for model training**
- **Improving diverse groups' representation during model training (e.g. SFT, RL)**
- **Changes to model architecture, systems**
- **Specific fairness/bias benchmarking**
- **More diverse benchmarks**
- **For specific applications: human oversight!**

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What are solutions? What to do about it?

- Changes to data used for model training **(hard)**
- Improving diverse groups' representation during model training (e.g. SFT, RL) **(hard)**
- Changes to model architecture, systems **(hard)**
- Specific fairness/bias benchmarking
- More diverse benchmarks
- For specific applications: human oversight!

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Quick Summary

Alignment: Ensuring an AI system's goals and outputs match human values, fairness, and safety

Hard to do!

AI fairness research is important

(safety: later in the course)

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Overall Summary

- Bias = disproportionate weight for/against an idea or a group.
- AI bias shows in what a model says, what it shows, and what it leaves out.
- Chatbots lean Western, regardless of language used.
- Bias enters at every stage: data, pre-training, SFT, RL, and the system prompt.
- Bias can be deliberate: creators intentionally steer the model.
- Harms take several forms, from stereotyping to unequal service and lost opportunity.
- Alignment is the idea of aligning AI with human values including fairness
- … But fixing the models is hard

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